

# Post-relaxation sanity checks for ML-relaxed adsorbates

codes/files/adsorbates.py

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## Why

codes/files/adsorbates.py generates adsorbate-on-slab structures and relaxes them with a machine-learning interatomic potential (MACE, uPET, MatterSim, UMA, ...). ML relaxers can extrapolate into regions they were never trained on and return *converged* geometries that are physically wrong. Because the energy is then stored under a label like \*COOH, a bad relaxation silently corrupts the CHE free-energy ladder downstream.

Composition is preserved by relaxation, so composition checks are useless. The failure modes that matter are geometric / topological:

1. **Dissociation / fragmentation** – \*COOH loses its H, \*OOH breaks its O-O bond. The recorded energy no longer belongs to the intended species. This is the dominant silent corruptor.
2. **Desorption** – the adsorbate floats off the slab; the “adsorption energy” is really  $E(\text{gas molecule}) + E(\text{slab})$ .
3. **Explosion / atom overlap** – the model drives atoms on top of each other or blows the structure apart.
4. **Isomerisation** – still intact and bound, but rearranged to a different connectivity than the label implies (\*OCHO vs \*COOH).
5. **Slab reconstruction** – the slab itself melts or ejects atoms, so the clean-slab reference no longer cancels.

## The check, in layers

Validation runs in `validate_relaxed_adsorbate()` after each adsorbate relaxes. Cheap filters run first and bail early. Layers 0-2 are **on by default**; layers 3-4 are **opt-in**.

| Layer                          | Catches                           | Default | Cost    |
|--------------------------------|-----------------------------------|---------|---------|
| 0 finite energy + atom overlap | NaN/inf energy, explosion/overlap | on      | trivial |
| 1 surface binding              | desorption                        | on      | cheap   |
| 2 molecular identity (graph)   | dissociation, isomerisation       | on      | cheap   |
| 3 slab integrity               | slab reconstruction               | off     | cheap   |
| 4 energy outliers (ensemble)   | subtle ML failures                | off     | medium  |

### Layer 0 – finite energy and overlap

The relaxed energy must be finite (not NaN/inf), and `has_reasonable_distances` (the same routine used to screen initial placements) must pass on the relaxed geometry: no pair of light atoms closer than  $0.5 * (r_{\text{cov}_i} + r_{\text{cov}_j})$ .

### Layer 1 – surface binding

At least one adsorbate atom must lie within `bind_tol * (r_cov_i + r_cov_j)` of at least one slab atom, using minimum-image (PBC-aware) distances. If nothing is within bonding range, the adsorbate has desorbed and the adsorption energy is meaningless. Default `bind_tol = 1.25`.

## Layer 2 – molecular identity

The intended species is known from `ads.properties['adsorbate']`, so we compare the relaxed adsorbate's intramolecular bond graph to a reference graph for that species:

- **2a Fragmentation:** the relaxed adsorbate must be a single connected component (every adsorbate in the library is one connected molecule). A fragment that breaks off creates a second component. This is the robust core check and depends only on the relaxed geometry.
- **2b Isomerisation:** the relaxed graph must be isomorphic (element-aware, via `networkx.is_isomorphic`) to the reference graph – catching rearrangements that keep the molecule in one piece.

Bonds are covalent-radii based: an edge connects  $i, j$  when their separation is within `graph_tol * (r_cov_i + r_cov_j)`. The relaxed graph uses minimum-image distances so it is robust to a molecule wrapping across a cell boundary.

### Calibrating `graph_tol`

The reference graphs are built from the (crude, unrelaxed) library geometries in `adsorbates.py`. The tolerance has to be loose enough that every reference is a single connected component, but tight enough not to invent spurious 1,3 bonds. Scanning all 38 bundled adsorbates:

- at 1.20, `*CHOH` is disconnected (its placeholder C-H sits at 1.295 Å);
- at 1.25, **all 38 references are connected** and no spurious bonds appear;
- spurious 1,3 bonds first appear at  $\geq 1.30$  (in `*OCCO`, `*ONNO`).

So the default is `graph_tol = 1.25`, on the plateau. Both the reference and the relaxed adsorbate use the same value (constant `_ADSORBATE_BOND_TOL`), which is what makes the comparison meaningful.

## Layer 3 – slab integrity (opt-in)

When `check_slab_integrity=True`, the slab atoms of the relaxed structure are compared to the clean relaxed slab (already computed in the same loop). If the maximum minimum-image displacement of any slab atom exceeds `slab_max_disp` (default 1.5 Å), the slab has reconstructed and the structure is rejected. The threshold is deliberately loose so that normal adsorption-induced relaxation ( $\sim 0.3$ - $0.5$  Å) does not trip it.

## Layer 4 – energy outliers (opt-in, ensemble)

When `check_energy_outliers=True`, after all sites on a slab are relaxed, the adsorbate energies are pooled **per species across sites**. A set is rejected if any of its adsorbates deviates from the species median by more than `energy_mad_factor` robust standard deviations ( $1.4826 * MAD$ , default factor 5). This catches subtle ML failures that pass every geometric check but give an absurd energy. It needs at least 4 sites per species to form a band.

## How it is wired in

- **Identifying the adsorbate atoms.** `generate_adsorbed_structures` appends the adsorbate after the slab, so the adsorbate is the last `n_ads` atoms. `n_ads` is read from the reference graph's node count. As a safety net, the validator checks that the last `n_ads` atoms have the expected element multiset; if not, it skips layers 1-3 (returns ok) rather than risk a false rejection from a bad index guess.
- **Reference graphs.** `generate_adsorbed_structures` returns `expected_graphs` (a dict `{adsorbate_name: networkx.Graph}`) built once per run from the reference Molecule geometries (with the X binding dummy stripped).
- **Hook point.** In `run_relaxation`, immediately after the convergence check and energy read, a failing structure is treated exactly like a relaxation failure: its set is dropped and the reason recorded. The all-or-nothing set logic is unchanged.

### `rejected.json`

Every run writes `rejected.json` (retrieved alongside `output.json`, `total.txt`, `failed.txt`) – a list of records:

```
[
  {"adsorbate": "*COOH", "site_type": "bridge",
   "ads_coord": [...], "repeat": [1, 1, 1],
   "reason": "dissociation (adsorbate fragmented)"}
]
```

It is an empty list when nothing was rejected. Nothing is ever dropped silently – if numbers look thin, read `rejected.json` to see why.

## CLI flags

`adsorbates.py` (the runtime script) accepts:

| Flag                                 | Effect                            |
|--------------------------------------|-----------------------------------|
| <code>--no-validate</code>           | turn OFF layers 0-2 (default: on) |
| <code>--check-slab-integrity</code>  | turn ON layer 3                   |
| <code>--check-energy-outliers</code> | turn ON layer 4                   |

The defaults (layers 0-2 on, 3-4 off) apply automatically through the AiiDA workchains with no further wiring, because `run_relaxation` defaults to `validate_adsorbates=True`.

## Tolerances and tuning

| Parameter                      | Default | Layer | Meaning                                               |
|--------------------------------|---------|-------|-------------------------------------------------------|
| <code>bind_tol</code>          | 1.25    | 1     | bond cutoff factor for adsorbate-slab binding         |
| <code>graph_tol</code>         | 1.25    | 2     | bond cutoff factor for the identity graph             |
| <code>slab_max_disp</code>     | 1.5     | 3     | max slab-atom displacement (Å) before “reconstructed” |
| <code>energy_mad_factor</code> | 5.0     | 4     | robust-sigma band half-width for outliers             |

Start lenient: the goal is to catch *garbage*, not to enforce tight geometry. Tighten only if junk slips through.

## Caveats

- **Metal bonding** is poorly described by covalent radii. That is why layer 2 graphs only the adsorbate’s *internal* (C/H/N/O/Cl) bonds, and layer 1 treats surface binding as a looser distance test rather than a graph edge.
- **Bidentate / multi-atom binding** is fine: the internal graph (e.g. the two C-O and one C-H bonds of `*OCHO`) is unchanged whether the molecule binds through one atom or two.
- **Crude reference geometries.** A few polyatomic coupling intermediates (`*OCCO`, `*ONNO`, `*CCHO`) have placeholder geometries with borderline bonds. If layer 2b false-rejects one of these, it is logged in `rejected.json` with a clear reason and can be tuned via `graph_tol` – the fragmentation check (2a) is unaffected and still catches dissociation.
- **Intended water release etc.** happens *across* pathway steps as separate species, never within a single adsorbate relaxation, so “intact” is the correct per-structure criterion.
- **Gas-phase references** are not validated here (out of scope); a future extension could apply the same identity check to the relaxed molecular\_references to guard against a reference molecule dissociating.

## Extending

To add a check, write it as another early-return branch in `validate_relaxed_adsorbate` (per-structure) or as a post-pass like `_flag_energy_outliers` (ensemble), gate it behind a `run_relaxation` flag if it is not cheap-and-safe enough to be on by default, and make sure failures land in `rejected.json` with a descriptive reason.